

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION WITH OPENCV
COURSE CODE:CSA0204

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- Viola-Jones: fast but less accurate (78%)
- YOLO: moderate speed, good accuracy (92%)
- Deep Learning: high accuracy (95%) but slow

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

2. Robust Detection under Environmental Variations

Detection accuracy across environments:

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

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Deep Learning	95%	85%	80%

Compare the robustness of each method and evaluate which technique provides the best overall performance across varying conditions. Justify trade-offs.

3. Optical Flow Reliability vs Noise Sensitivity

Two optical flow methods show:

- Method A: stable vectors but misses small motions
- Method B: detects fine motion but noisy vectors

Compare both methods and evaluate which is more suitable for motion tracking in dynamic scenes. Justify based on application requirements.

4. Background Modeling Strategy (GMM)

A GMM-based system shows:

- High false positives in dynamic background
- Missed detections in static background

Compare GMM performance across scenarios and evaluate whether GMM alone is sufficient. Propose and justify improvements.

5. Motion Estimation vs Computational Cost

Two methods:

- Method A: low error (5%), high computation
- Method B: moderate error (10%), low computation

Compare the methods and evaluate their suitability for real-time vs offline applications. Justify your decision.

6. Structure from Motion Consistency

Two datasets produce:

- Dataset A: accurate depth but requires high-quality images
- Dataset B: inconsistent depth but works on low-quality images

Compare the datasets and evaluate which setup is more practical for real-world deployment. Justify your reasoning.

7. Stereo Vision vs Motion Estimation

Depth estimation results:

- Stereo Vision: accurate but requires dual cameras
- Motion Estimation: less accurate but works with single camera

Compare both approaches and evaluate which is preferable under hardware constraints. Justify your choice.

8. Model Generalization vs Overfitting

Performance:

Model Training Testing

A	98%	70%
B	90%	88%

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

9. YOLO Performance across Scales

Detection results:

- Large objects: 95%
- Medium objects: 85%
- Small objects: 60%

Compare performance across scales and evaluate whether YOLO alone is sufficient for multi-scale detection tasks. Justify improvements.

10. Motion Model Selection for Animation

Two models:

- Model A: smooth but physically inaccurate
- Model B: realistic but computationally intensive

Compare the models and evaluate which should be used for animation systems under different constraints (real-time vs cinematic). Justify your decision.

11. Detection Model Comparison under Resource Constraints

Performance:

	Model Accuracy	Latency
A	85%	40 ms
B	92%	90 ms
C	90%	60 ms

Compare the models and evaluate which is optimal for real-time systems with moderate accuracy requirements. Justify your decision.

12. Robustness vs Speed Trade-off

Performance:

Method	Accuracy	Robustness	Speed
A	88%	High	Slow
B	85%	Moderate	Fast
C	90%	Low	Moderate

Compare the methods and evaluate the best choice for a balanced system. Justify trade-offs.

13. Optical Flow Stability Analysis

Results:

Method	Accuracy	Noise Sensitivity
A	High	High
B	Moderate	Low

Compare both methods and evaluate suitability for real-time tracking in noisy environments.

14. Motion Estimation Accuracy vs Efficiency

Results:

Method	Error	Computation
A	4%	High
B	9%	Low
C	6%	Moderate

Compare methods and evaluate the best approach for embedded systems.

15. GMM Performance in Different Conditions

Results:

Scenario	Accuracy
Static	92%
Dynamic	68%

Compare performance and evaluate whether GMM is reliable for all environments. Suggest improvements.

16. Object Detection under Scale Variation

Results:

Object Size	Accuracy
Large	94%
Medium	80%
Small	55%

Compare detection performance and evaluate system limitations. Justify improvements.

17. Stereo vs SfM Accuracy Comparison

Results:

Method	Accuracy	Hardware
Stereo	High	Dual camera
SfM	Moderate	Single camera

Compare and evaluate which is more practical under hardware limitations.

18. Detection Model Generalization

Results:

Model	Training	Testing
A	97%	75%
B	90%	88%

Compare and evaluate which model is better for deployment.

19. Motion Tracking Accuracy Comparison

Results:

Method	Accuracy	Stability
A	High	Low
B	Moderate	High

Compare and evaluate which is suitable for long-term tracking.

20. Surveillance Model Selection

Results:

Model	Accuracy	Latency
A	82%	30 ms
B	90%	70 ms
C	95%	120 ms

Compare and evaluate under strict latency constraints.

21. Noise Impact on Detection

Results:

Noise Level	Accuracy
Low	92%
Medium	80%
High	60%

Compare performance and evaluate system robustness.

22. Depth Estimation Error Analysis

Results:

Method	Error
A	0.5 m
B	1.2 m
C	0.8 m

Compare and evaluate accuracy for real-world applications.

23. Multi-Model Comparison

Results:

Model	Accuracy	Speed	Robustness
A	88%	Fast	Low
B	92%	Moderate	High

C	95%	Slow	High
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Compare and evaluate the best overall model.

24. Motion Detection under Crowd Density

Results:

Method	Sparse Crowd	Dense Crowd
Optical Flow	High	Low
Motion Models	Moderate	High

Compare and evaluate best approach.

25. Model Efficiency vs Accuracy

Results:

Model	Accuracy	Computation
A	90%	High
B	85%	Low
C	88%	Moderate

Compare and evaluate for edge devices.

26. Feature Detection under Illumination Changes

Results:

Method	Normal	Low Light
A	90%	60%
B	85%	75%

Compare and evaluate robustness.

27. Video Processing Pipeline Comparison

Results:

Pipeline	Accuracy	Speed
A	88%	Fast
B	92%	Moderate
C	95%	Slow

Compare and evaluate suitability.

28. Motion Model Evaluation

Results:

Model	Realism	Speed
A	High	Low
B	Moderate	High

Compare and evaluate based on application context.

29. Detection under Occlusion

Results:

Method	Accuracy
A	70%
B	85%
C	90%

Compare and evaluate robustness under occlusion.

30. System Performance under Constraints

Results:

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Compare and evaluate the best system considering all constraints.

Code of Conduct:

I BHAVYA PS9192524017 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Bhavya PS

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions

Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand
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1. Model Efficiency vs Accuracy

Given:

- Model A → Accuracy: 90%, Computation: High
- Model B → Accuracy: 85%, Computation: Low
- Model C → Accuracy: 88%, Computation: Moderate

Model	Accuracy	Computation
A	90%	High
B	85%	Low
C	88%	Moderate

Comparison:

- Model A has the highest accuracy but requires high computation.
- Model B requires the least computation but has the lowest accuracy.
- Model C has good accuracy with moderate computation.

Analysis:

- Edge devices have limited power and processing resources.
- Model A may consume more power and processing time.
- Model B is efficient but may give less accurate results.
- Model C provides a better balance between accuracy and computation.

Choice:

Model C

Conclusion:

Model C is the most suitable for edge devices because it provides good accuracy with moderate computational requirements.

2. Optical Flow Reliability vs Noise Sensitivity

Given:

- Method A → Stable vectors, but misses small motions.
- Method B → Detects fine motion, but produces noisy vectors.

Comparison:

- Method A is more stable and reliable.
- Method B is more sensitive and can detect small movements.
- However, noise in Method B can cause incorrect motion detection.

Analysis:

- Dynamic scenes require stable motion tracking.
- Method A is better when reliable tracking is more important.
- Method B is useful when detecting very small movements is the main requirement.

Choice:

Method A for general dynamic-scene tracking

Conclusion:

Method A is more suitable for reliable motion tracking because its vectors are stable. Method B is preferable only when fine-motion detection is more important.

3. Stereo Vision vs Motion Estimation

Given:

- Stereo Vision → High accuracy, requires two cameras.
- Motion Estimation → Lower accuracy, requires one camera.

Comparison:

- Stereo Vision provides more accurate depth information.
- It requires additional camera hardware and calibration.
- Motion Estimation uses only one camera.
- It is cheaper and simpler but less accurate.

Analysis:

- Hardware-constrained systems need fewer components.
- Using two cameras increases cost and system complexity.
- A single-camera solution is more practical when hardware is limited.

Choice:

Motion Estimation

Conclusion:

Under hardware constraints, Motion Estimation is preferable because it requires only one camera and reduces cost and complexity, even though its accuracy is lower.

4. Robustness vs Speed Trade-off

Given:

Method	Accuracy	Robustness	Speed
A	88%	High	Slow
B	85%	Moderate	Fast
C	90%	Low	Moderate

Comparison:

- Method A → High robustness, good accuracy, but slow.
- Method B → Fastest, but has lower accuracy.
- Method C → Highest accuracy, but low robustness.

Analysis:

- A reliable system needs good accuracy and robustness.
- Method B is useful for real-time applications where speed is important.
- Method C may fail when conditions change because of its low robustness.
- Method A gives more reliable performance despite being slower.

Choice:

Method A

Conclusion:

Method A is the best balanced choice because it provides high robustness and good accuracy. Its main disadvantage is slower processing.

5. System Performance under Constraints

Given:

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Comparison:

- Model A → Fastest and cheapest, but lowest accuracy.
- Model B → Good accuracy, moderate latency, and moderate cost.
- Model C → Highest accuracy, but slowest and most expensive.

Analysis:

- Model A is suitable when speed and low cost are priorities.
- Model C is suitable when maximum accuracy is more important than cost and latency.
- Model B provides a good balance between all three factors.

Choice:

Model B

Conclusion:

Model B is the best overall choice because it provides 92% accuracy with reasonable latency and moderate cost. It offers the best balance for a practical system.